$\boxed{4.1}$

$$p_i = -i\hbar \frac{\partial}{\partial r_i}$$

(a)

$$[r_j, p_i]f = r_j \hat{p}_i f - \hat{p}_i r_j f$$

$$= -i\hbar \left[r_j \frac{\partial f}{\partial r_i} - \frac{\partial}{\partial r_i} r_j f \right]$$

but $\rightarrow \dfrac{\partial r_j}{\partial r_i} = \delta_{ij}$

$$= -i\hbar \left[\cancel{r_j \frac{\partial f}{\partial r_i}} - \delta_{ij} f - \cancel{r_j \frac{\partial f}{\partial r_i}} \right]$$

$$\Rightarrow [r_j, p_i]f = i\hbar \delta_{ij} f \Rightarrow \boxed{[r_j, p_i] = i\hbar \delta_{ij}}$$

Notice $[A,B] = -[B,A]$, thus $[r_i, p_j] = -[p_j, r_i]$

$[r_i, r_j] = r_i r_j - r_j r_i = 0 \rightarrow$ Coordinates Commute with each other

$$\boxed{[r_i, r_j] = 0}$$

$$[p_i, p_j]f = (p_i p_j - p_j p_i)f = (-i\hbar)^2 \left[\frac{\partial}{\partial r_i}\frac{\partial}{\partial r_j} - \frac{\partial}{\partial r_j}\frac{\partial}{\partial r_i} \right] f$$

since $\rightarrow \dfrac{\partial}{\partial r_i}\dfrac{\partial}{\partial r_j} = \dfrac{\partial}{\partial r_j}\dfrac{\partial}{\partial r_i}$

$$\Rightarrow \boxed{[p_i, p_j] = 0}$$

(b) ~~For verification~~

$$\frac{d}{dt}\langle \vec{r} \rangle = \frac{\langle \vec{p} \rangle}{m}$$

since → $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

and $\vec{p} = p_x\hat{i} + p_y\hat{j} + p_z\hat{k}$, where $p_x = -i\hbar\frac{\partial}{\partial x}$, $p_y = -i\hbar\frac{\partial}{\partial y}$, $p_z = -i\hbar\frac{\partial}{\partial z}$

to prove : $$\frac{d\langle \vec{r} \rangle}{dt} = \frac{d\langle x \rangle}{dt}\hat{i} + \frac{d\langle y \rangle}{dt}\hat{j} + \frac{d\langle z \rangle}{dt}\hat{k}$$

$$= \frac{\langle \vec{p} \rangle}{m} = \frac{1}{m}\left[\langle p_x \rangle \hat{i} + \langle p_y \rangle \hat{j} + \langle p_z \rangle \hat{k}\right]$$

so let us only prove → $\frac{d\langle x \rangle}{dt} = \frac{1}{m}\langle p_x \rangle$, rest will follow similarly.

$$\frac{d}{dt}\langle x \rangle = \frac{i}{\hbar}\left\langle [\hat{H}, \hat{x}] \right\rangle + \cancel{\left\langle \frac{\partial \hat{x}}{\partial t} \right\rangle}^{\,0} \longrightarrow \text{Eqn } 3.71$$

$$= \frac{i}{\hbar}\left\langle \left[\frac{\hat{p}_x^2}{2m} + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m} + V(x,y,z), x\right] \right\rangle$$

Note: $[A+B, C] = [A, C] + [B, C]$

Also → $[V, x] = 0$ ← coordinates commute

$[\hat{p}_z^2, \hat{x}] = [\hat{p}_y^2, \hat{x}] = 0$ ← since $[\hat{p}_y, x] = [\hat{p}_z, x] = 0$

and ~~we~~ $[AB, C] = A[B, C] + [A, C]B$

where $A = \hat{p}_z$, $B = \hat{p}_z$

or $A = \hat{p}_y$, $B = \hat{p}_y$

Hence remaining → $\frac{d\langle x \rangle}{dt} = \frac{i}{\hbar}\left\langle \left[\frac{\hat{p}_x^2}{2m}, \hat{x}\right] \right\rangle = \frac{\langle \hat{p}_x \rangle}{m}$ ← HW # 8 P3.17 (c)

rest $\rightarrow$ $\dfrac{d\langle y\rangle}{dt} = \dfrac{\langle \hat{p}_y\rangle}{m}$ and $\dfrac{d\langle \hat{z}\rangle}{dt} = \dfrac{\langle \hat{p}_z\rangle}{m}$, identical approach.

to prove: $\dfrac{d}{dt}\langle \vec{p}\rangle = \langle -\vec{\nabla} V\rangle$

Note $\vec{\nabla} = \dfrac{\partial}{\partial x}\hat{i} + \dfrac{\partial}{\partial y}\hat{j} + \dfrac{\partial}{\partial z}\hat{k}$

and $\vec{\nabla} V = \dfrac{\partial V}{\partial x}\hat{i} + \dfrac{\partial V}{\partial y}\hat{j} + \dfrac{\partial V}{\partial z}\hat{k}$; where $V = V(x, y, z)$

again it suffices to prove $\rightarrow$

$$\frac{d\langle p_x\rangle}{dt} = -\left\langle \frac{\partial V}{\partial x}\right\rangle$$

$$\frac{d\langle p_x\rangle}{dt} = \frac{i}{\hbar}\left\langle \left[\frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m} + V,\ p_x\right]\right\rangle + \cancel{\left\langle \frac{\partial p_x}{\partial t}\right\rangle}^{0} \rightarrow \boxed{3.7}$$

$$= \frac{i}{\hbar}\langle [V, p_x]\rangle$$

$$[V, p_x] f = \left[V, -i\hbar\frac{\partial}{\partial x}\right] f$$

$$= -i\hbar\left[V\frac{\partial}{\partial x} - \frac{\partial}{\partial x}V\right] f$$

$$= -i\hbar\, V\frac{\partial f}{\partial x} - \frac{\partial (Vf)}{\partial x}$$

$$= -i\hbar\left[\cancel{V\frac{\partial f}{\partial x}} - \cancel{V\frac{\partial f}{\partial x}} - \left(\frac{\partial V}{\partial x}\right) f\right]$$

$$\Rightarrow [V, p_x] f = i\hbar\left(\frac{\partial V}{\partial x}\right) f$$

$$\Rightarrow \frac{d\langle p_x\rangle}{dt} = -\left\langle \frac{\partial V}{\partial x}\right\rangle \xrightarrow{\text{similarly}} \frac{d\langle p_y\rangle}{dt} = -\frac{\partial V}{\partial y} \text{ and } \frac{d\langle p_z\rangle}{dt} = -\frac{\partial V}{\partial z}$$

(c) Use 3.62

$$\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2$$

$$\sigma_{r_i}^2 \sigma_{p_j}^2 \geq \left(\frac{1}{2i} \langle [\hat{r}_i, \hat{p}_j] \rangle \right)^2 = \delta_{ij} \frac{\hbar^2}{4}$$

$$\Rightarrow \sigma_{r_i} \sigma_{p_j} \geq \delta_{ij} \frac{\hbar}{2} \quad ; \quad \text{Note} \rightarrow \delta_{ij}^2 = \delta_{ij}$$

similarly $\rightarrow \sigma_{r_i} \sigma_{r_j} \geq 0$

and $\sigma_{p_i} \sigma_{p_j} \geq 0$.

$\boxed{4.2}$

$$-\frac{\hbar^2}{2m} \left[\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right] \psi = \varepsilon \psi$$

$$\psi = f(x)\, g(y)\, s(z)$$

$$\Rightarrow -\frac{\hbar^2}{2m} \underbrace{\left[\frac{d^2 f(x)}{f\, dx^2} + \frac{d^2 g(y)}{g(y)\, dy^2} + \frac{1}{s(z)} \frac{d^2 s(z)}{dz^2} \right]}_{\text{All three terms need to be constant,}} = \varepsilon$$

thus $\rightarrow$ $$-\frac{\hbar^2}{2m} \frac{d^2 f}{f\, dx^2} = \alpha$$

$$-\frac{\hbar^2}{2m} \frac{d^2 g}{g\, dx^2} = \beta$$

$$-\frac{\hbar^2}{2m} \frac{d^2 s(z)}{s\, dz^2} = \gamma$$

or $\rightarrow \quad \mathcal{E} = \alpha + \beta + \gamma > 0$

Because of the separation of variables it is clear that on all $\alpha > 0$

$\beta > 0$

$\gamma > 0$

thus $\rightarrow \quad f = A_x \cos\left[\sqrt{\frac{2m}{\hbar^2}\alpha}\; x\right] + B_x \sin\left[\sqrt{\frac{2m\alpha}{\hbar^2}}\; x\right]$

from Boundry conditions $\rightarrow f[0] = f[a] = 0$

$\Rightarrow A_x = 0$

and $\sqrt{\frac{2m\alpha}{\hbar^2}}\, a = \pi n_x \quad ; \quad n_x = 0, 1, 2, \ldots$

thus $\rightarrow \quad \alpha = \hbar^2 \frac{\pi^2 n_x^2}{2ma^2}$

similarly $\rightarrow \quad \beta = \frac{\hbar^2 \pi^2 n_y^2}{2ma^2} \; ; \quad n_y = 0, 1, 2 \ldots$

$\gamma = \frac{\hbar^2 \pi^2 n_z^2}{2ma^2} \; ; \quad n_z = 0, 1, 2, \ldots$

So $\rightarrow \quad \mathcal{E} = \frac{\pi^2 \hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$

and state $\quad \psi = f_{n_x}(x)\, g_{n_y}(y)\, S_{n_z}(z)$

$$= \left[\sqrt{\frac{2}{a}}\right]^{3/2} \sin\left[\frac{n_x \pi}{a} x\right] \sin\left[\frac{n_y \pi}{a} y\right] \sin\left[\frac{n_z \pi}{a} z\right]$$

3D - Normalization Constant

(b)

n_x	n_y	n_z	$n_x + n_y + n_z$	ε	degeneracy
1	1	1	3	$\varepsilon_1 = \frac{3\pi^2\hbar^2}{2ma^2}$	1
1	2	1	6	$\varepsilon_2 = \frac{6\pi^2\hbar^2}{2ma^2}$	3
2	1	1	6		
1	1	2	6		
1	2	2	9	$\varepsilon_3 = \frac{9\pi^2\hbar^2}{2ma^2}$	3
2	1	2	9		
2	2	1	9		
1	1	3	11	$\varepsilon_4 = \frac{11\pi^2\hbar^2}{2ma^2}$	3
3	1	1	"		
1	3	1	"		
2	2	2	12	$\varepsilon_5 = \frac{12\pi^2\hbar^2}{2ma^2}$	1
1	2	3	14	$\varepsilon_6 = \frac{14\pi^2\hbar^2}{2ma^2}$	6
3	1	2	"		
2	3	1	"		
3	2	1	"		
1	3	2	"		
2	1	3	"		

(c) the $E_{14} \rightarrow$ comes from the combination

$$3^2+3^2+3^2 = 27 \longrightarrow 3, 3, 3 \longrightarrow d = 1$$

and $5^2+1^2+1^2 = 27 \rightarrow 5, 1, 1 \longrightarrow d = 3$

So total degeneracy $d = 1+3 = 4$

whereas for all energies $< E_{14}$

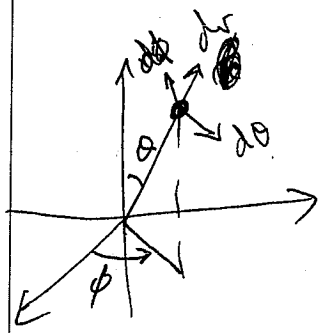
the degeneracies are either $\rightarrow d = 1$ or 3 or 6 only!

So for E_{14} the number of degeneracy cannot be obtained by only combinatorial arguments!

Curvilinear Coordinates

(u, v, w) (x, y, z) (r, θ, ϕ) (s, ϕ, z)

cartesian spherical cylindrical



displacement vector

$$d\vec{l} = (u+du,\ v+dv,\ w+dw) - (u, v, w)$$

$$= f\,du\,\hat{u} + g\,dv\,\hat{v} + h\,dw\,\hat{w}$$

$(\hat{u}, \hat{v}, \hat{w})$ = orthogonal unit vectors

Scalar function changes by

$$dt = \frac{\partial t}{\partial u}du + \frac{\partial t}{\partial v}dv + \frac{\partial t}{\partial w}dw$$

$$dt = \vec{\nabla} t \cdot d\vec{l} = (\nabla t)_u\, f\,du + (\nabla t)_v\, g\,dv + (\nabla t)_w\, h\,dw$$

where we defined
$$(\nabla t)_u = \frac{1}{f}\frac{\partial t}{\partial u}$$
$$(\nabla t)_v = \frac{1}{g}\frac{\partial t}{\partial v}$$
$$(\nabla t)_w = \frac{1}{h}\frac{\partial t}{\partial w}$$

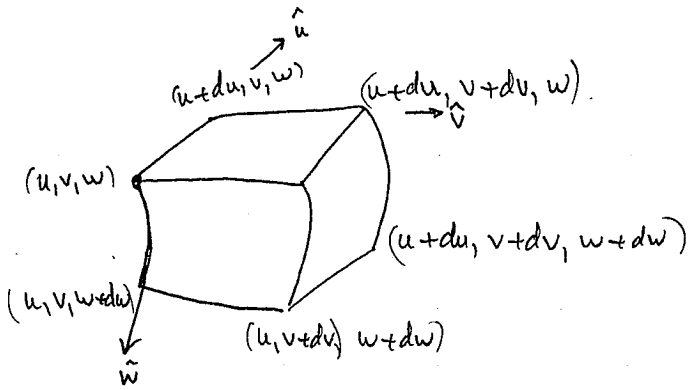
$\Rightarrow$ the gradient is

$$\vec{\nabla} t = \frac{1}{f}\frac{\partial t}{\partial u}\hat{u} + \frac{1}{g}\frac{\partial t}{\partial v}\hat{v} + \frac{1}{h}\frac{\partial t}{\partial w}\hat{w}$$

System	u v w	f g h
Cartesian	x y z	1 1 1
Spherical	r θ φ	1 r r sin θ
Cylindrical	s φ z	1 s 1

Use $\oint \vec{A}\cdot d\vec{a} = \int \vec{\nabla}\cdot\vec{A}\, d\tau$

$$\vec{A}(u,v,w) = A_u\,\hat{u} + A_v\,\hat{v} + A_w\,\hat{w}$$



$$d\tau = f g h\, du\, dv\, dw$$
$$= dl_u\, dl_v\, dl_w$$

$$\begin{cases} dl_u = f\,du \\ dl_v = g\,dv \\ dl_w = h\,dw \end{cases}$$

<u>Front face</u> $d\vec{a} = -\hat{u}\, dl_w\, dl_v = -gh\, dv\, dw\, \hat{u}$

$$\vec{A}\cdot d\vec{a} = -(gh A_u)\, dv\, dw$$

<u>Back face</u> $\vec{A}\cdot d\vec{a} = +\underbrace{(gh A_u)}_{\text{evaluated at } \underline{u+du}}\, dv\, dw$

but $F(u+du) - F(u) = \dfrac{dF}{du}\, du$

$$\Rightarrow \underline{\text{front + back}} = \frac{\partial}{\partial u}(gh A_u)\underbrace{dv\, dw\, du}_{\frac{d\tau}{fgh}} = \frac{1}{fgh}\frac{\partial}{\partial u}(gh A_u)\, d\tau$$

<u>left</u> $d\vec{a} = -\hat{v}\, dl_u\, dl_w = -(fh)\, du\, dw\, \hat{v}$

$$\vec{A}\cdot d\vec{a} = -(fh A_v)\, du\, dw$$

$$\underline{\text{right + left}} = \frac{\partial}{\partial v}(fh A_v)\frac{d\tau}{fgh}$$

$$\underline{\text{Top + Bottom}} = \frac{\partial}{\partial w}(gf A_w)\frac{d\tau}{fgh}$$

$$\Rightarrow \oint \vec{A}\cdot d\vec{a} = \frac{1}{fgh}\left\{\frac{\partial}{\partial u}(ghA_u) + \frac{\partial}{\partial v}(fhA_v) + \frac{\partial}{\partial w}(fgA_w)\right\} d\tau$$

$$\Rightarrow \vec{\nabla}\cdot\vec{A} = \frac{1}{fgh}\left\{\frac{\partial}{\partial u}(ghA_u) + \frac{\partial}{\partial v}(fhA_v) + \frac{\partial}{\partial w}(fgA_w)\right\}$$

$\Rightarrow$ <u>Laplacian</u> $\nabla^2 t = \vec{\nabla}\cdot\vec{\nabla} t =$

$$= \frac{1}{fgh}\left[\frac{\partial}{\partial u}\left(\frac{gh}{f}\frac{\partial t}{\partial u}\right) + \frac{\partial}{\partial v}\left(\frac{fh}{g}\frac{\partial t}{\partial v}\right) + \frac{1}{\partial w}\left(\frac{fg}{h}\frac{\partial t}{\partial w}\right)\right]$$

<u>Spherical coordinates:</u>

$$\nabla^2 t = \frac{1}{r^2\sin\theta}\left[\left(\frac{\partial}{\partial r} r^2\sin\theta\frac{\partial t}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{r\sin\theta}{r}\frac{\partial t}{\partial\theta}\right)\right.$$

$$\left. + \frac{\partial}{\partial\varphi}\left(\frac{r}{r\sin\theta}\frac{\partial t}{\partial\varphi}\right)\right] =$$

$$= \frac{1}{r^2}\left(\frac{\partial}{\partial r} r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial t}{\partial\theta}\right)$$

$$+ \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial\varphi^2}$$

$$\Rightarrow \nabla^2 = \frac{1}{r^2}\left(\frac{\partial}{\partial r} r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$

<u>Displacement</u> $d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$

$$d\tau = r^2 \sin\theta\, dr\, d\theta\, d\phi \qquad \vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\right) = 4\pi\delta^3(\vec{r})$$

$$f = 1 \quad g = r \quad h = r\sin\theta \qquad \nabla^2 \frac{1}{r} = -4\pi\delta^3(\vec{r})$$

<u>Result</u> Then $\underbrace{\vec{\nabla}^2 t}_{\vec{\nabla}\cdot\vec{\nabla} t} = \frac{1}{fgh}\left[\frac{\partial}{\partial u}\left(\frac{gh}{f}\frac{\partial t}{\partial u}\right) + \frac{\partial}{\partial v}\left(\frac{fh}{g}\frac{\partial t}{\partial v}\right) + \frac{\partial}{\partial w}\left(\frac{fg}{h}\frac{\partial t}{\partial w}\right)\right]$

$$\nabla^2 t = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial t}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{r\sin\theta}{r}\frac{\partial t}{\partial\theta}\right) + \frac{\partial}{\partial\phi}\left(\frac{r}{r\sin\theta}\frac{\partial t}{\partial\phi}\right)\right] =$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial t}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial\phi^2} \quad \text{for any } t.$$

$$\Rightarrow \nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$